Period/year	Development/ innovation	Developer/country	Key features/technological advancement

1.3.2 Breakthroughs in electronics and navigation

The rapid progress of drone technology in recent decades has been largely enabled by breakthroughs in miniaturized electronics, embedded sys-tems, and navigation technologies. The integration of lightweight sensors, microcontrollers, and efficient propulsion systems has drastically reduced size and cost while enhancing reliability and performance. The introduction of the global positioning system (GPS) revolutionized drone navigation, allowing for precise positioning, autonomous waypoint tracking, and stable flight operations. Inertial measurement units (IMUs), gyroscopes, accelerometers, and magnetometers further improved flight stability and control accuracy. Advances in wireless communication, particularly in real-time data transmission and remote telemetry, have facilitated long-range operations and seamless integration with ground control systems. Together, these innovations form the technological backbone of modern UAV platforms, enabling diverse civilian and defense applications.

1.3.3 Role of artificial intelligence and autonomy

Artificial intelligence (AI) has emerged as a defining force in the latest generation of drone technology, shifting systems from remote-controlled platforms to fully autonomous aerial entities. AI algorithms enable drones to process sensor data, recognize patterns, and make real-time decisions without human intervention. Machine learning enhances flight optimization, object detection, obstacle avoidance, and mission planning, allowing UAVs